Problem Solving with Volumes

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Problem-solving with volumes

What is problem-solving?

- **Problem-solving**, usually has two key features:
  - A question is given as a **real-life scenario**
    - eg. The volume of water in a swimming pool...
  - There is usually **more than one topic of maths** you will need in order to answer the question
    - eg. Volume and money

What are common problems that involve volume?

- **Volume** is a commonly used topic of **'real-world'** maths
  - For example, a carton of juice in the shape of a cuboid, a cylindrical tin and a triangular prism chocolate box all involve volume
- Typically, the 'real-world' scenarios also have a **cost**
  - A lot of volume problems also involve **calculations with money**

How do I solve problems involving volume?

- Often the 3D object in a question will **not** be a standard cuboid, cone, sphere, etc.
  - It will likely either be:
    - A **prism** (3D shape with the same cross-section running through it)
    - A **portion** or **fraction** of a standard shape (a hemisphere for example)
    - A **compound object** (an object made up of two or more standard 3D objects)
- If the object is a **prism**, recall that the **volume of a prism** is the **cross-sectional area × its length**
  - The cross-sectional area may be a **compound 2D shape**
    - For example, an L-shape, or a combination of a rectangle and a triangle

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

- A hemisphere is half a sphere
- A **frustum** is a **truncated** (chopped-off) **cone** or **pyramid**
  - The volume of a frustum will be the volume of the smaller cone or pyramid subtracted from the volume of the larger cone or pyramid

- If the object is a **compound object**, find the volumes of the **individual** standard 3D objects and **add** them together
- Problem solving questions could appear on either a **non-calculator paper** or a **calculator paper**

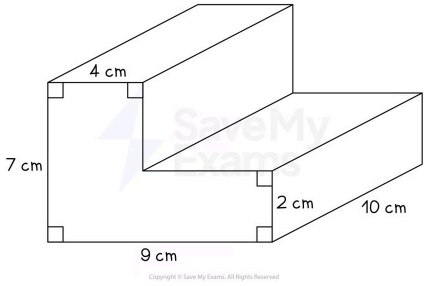
Examiner Tips and Tricks

Before you start calculating, make a quick note of your plan to tackle the question.

For example, "Find the area of the triangle and the rectangle, add together, multiply by the length".

Worked Example

The diagram shows a prism.



Work out the volume of the prism.

Answer:

The volume is the area of the cross section × length (10 cm)
Find the area by splitting into a 7 × 4 and a (9 - 4) × 2 rectangle (or a 9 × 2 and a (7 - 2) × 4 rectangle)

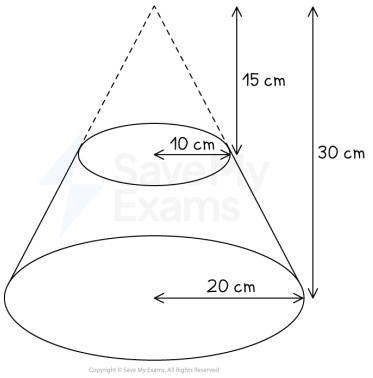
$$7 \times 4 + (9 - 4) \times 2 = 38 \text{ cm}^2$$

Find the volume (by multiplying 38 by 10)

$$38 \times 10$$

$$\mathbf{380 \text{ cm}^3}$$

The diagram shows a truncated cone (a frustum).
Using the given dimensions, find the volume of the frustum to 3 significant figures.



Answer:

To find the volume of the frustum, find the volume of the larger cone (30 cm tall, with a radius of 20 cm), and subtract the volume of the smaller cone (15 cm tall, with a radius of 10 cm)

Formula for the volume of a cone: $\frac{1}{3}\pi r^2 h$

Calculate the volume of the larger cone

$$V_L = \frac{1}{3} \times \pi \times 20^2 \times 30 = 4000\pi = 12\,566.37061\ldots \text{ cm}^3$$

Calculate the volume of the smaller cone

$$V_S = \frac{1}{3} \times \pi \times 10^2 \times 15 = 500\pi = 1\,570.796327\ldots \text{ cm}^3$$

Find the difference

$$V_L - V_S = 4000\pi - 500\pi = 3500\pi = 10\,995.57429\ldots \text{ cm}^3$$

Round to 3 significant figures

11 000 cm^3

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